

When should I suspect type 2 diabetes in an adult?

Last revised in March 2025

Suspect a diagnosis of type 2 diabetes if an adult presents with persistent hyperglycaemia that may be accompanied by clinical features.

- Possible clinical features of type 2 diabetes include:
 - Symptoms such as polydipsia, polyuria, blurred vision, unexplained weight loss, recurrent infections, and tiredness. Note: these may be mild or absent.
 - Signs such as acanthosis nigricans (a skin condition causing dark pigmentation of skin folds, typically the axillae, groin, and neck), which suggests insulin resistance.
 - The presence of risk factors (</topics/diabetes-type-2/background-information/risk-factors/>).
- Persistent hyperglycaemia is defined as:
 - HbA1c of 48 mmol/mol (6.5%) or more.
 - Fasting plasma glucose level of 7.0 mmol/L or more.
 - Random plasma glucose of 11.1 mmol/L or more in the presence of symptoms or signs of diabetes.
 - If the person is symptomatic, a single abnormal HbA1c or fasting plasma glucose level can be used, although repeat testing is sensible to confirm the diagnosis.
 - If the person is asymptomatic, do not diagnose diabetes on the basis of a single abnormal HbA1c or plasma glucose result. Arrange repeat testing, preferably with the same test, to confirm the diagnosis. If the repeat test result is normal, arrange to monitor the person for the development of diabetes, the frequency depending on clinical judgement.
 - Note: be aware that severe hyperglycaemia in people with acute infection, trauma, circulatory or other stress may be transitory and is not diagnostic of diabetes.
- **If the HbA1c result is difficult to interpret** (</topics/diabetes-type-2/diagnosis/diagnosis-in-adults/#interpreting-hba1c-results>):
 - Diagnose type 2 diabetes on the basis of a fasting plasma glucose level of 7.0 mmol/L or more.
 - If there is uncertainty about the use of HbA1c in the diagnosis or monitoring of type 2 diabetes, seek advice from the specialist diabetes team or clinical biochemistry.
- **When confirming the diagnosis, be aware type 2 diabetes is more likely if there are:**
 - No additional features of type 1 diabetes (such as rapid onset, often in childhood, insulin dependence, or ketoacidosis). See the CKS topic on [Diabetes – type 1](/topics/diabetes-type-1/) (</topics/diabetes-type-1/>) for more information.
 - No clinical features of other types of diabetes (</topics/diabetes-type-2/background-information/definition/>), such as monogenic diabetes or diabetes secondary to another condition or disease.

Interpreting HbA1c results

- **HbA1c should not be used to diagnose diabetes mellitus in the following groups:**
 - Children and young people less than 18 years of age.
 - Pregnant women or women who are 2 months postpartum.
 - People with symptoms of diabetes for less than 2 months.
 - People at high diabetes risk who are acutely ill.
 - People taking medication that may cause hyperglycaemia (for example long-term corticosteroid treatment).
 - People with acute pancreatic damage, including pancreatic surgery.
 - People with end-stage renal disease (ESRD). See the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](/topics/chronic-kidney-disease/) for more information.
- **HbA1c should be interpreted with caution in people with abnormal red blood cell turnover or abnormal haemoglobin type:**
 - Abnormal haemoglobin, such as haemoglobinopathy, including sickle cell trait. Black people (including those with potentially undiagnosed sickle cell trait) may need further evaluation when considering an HbA1c test result as the result may underestimate past glycaemia.
 - Severe anaemia (any cause). See the CKS topics on [Anaemia - iron deficiency \(/topics/anaemia-iron-deficiency/\)](/topics/anaemia-iron-deficiency/) and [Anaemia - B12 and folate deficiency \(/topics/anaemia-b12-folate-deficiency/\)](/topics/anaemia-b12-folate-deficiency/) for more information.
 - Altered red cell lifespan (for example post-splenectomy).
 - A recent blood transfusion.

[WHO, 2011 (</topics/diabetes-type-2/references/>); Kilpatrick and Atkin, 2014 (</topics/diabetes-type-2/references/>); Davies, 2018 (</topics/diabetes-type-2/references/>); Milne, 2020 (</topics/diabetes-type-2/references/>); NICE, 2020a (</topics/diabetes-type-2/references/>)]

Basis for recommendation

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